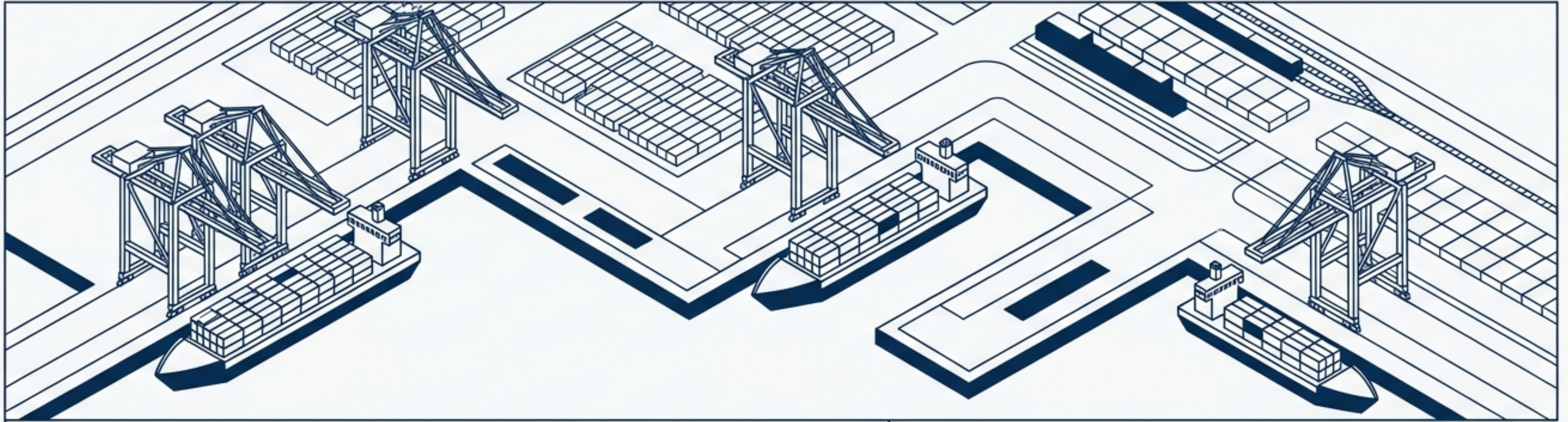




PortPilot AI

Explainable Port Congestion
Prediction & 72-Hour
Planning System

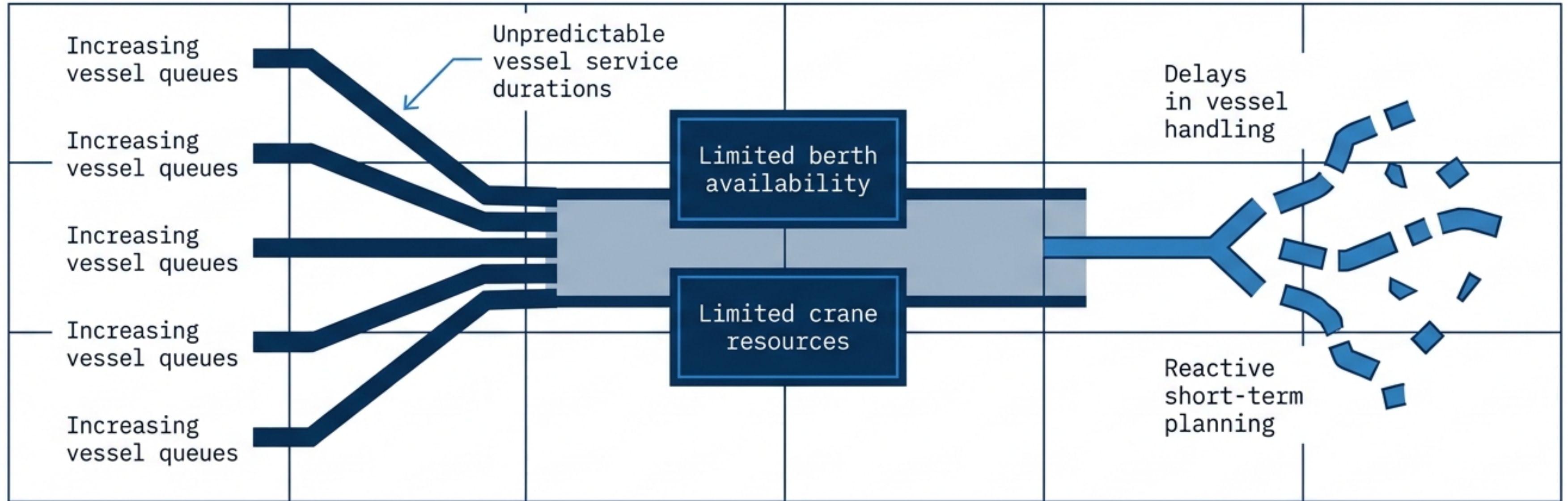
Team Name: Elite

Team Members: Harsh • Dax • Maulik • Deep

Track: AI

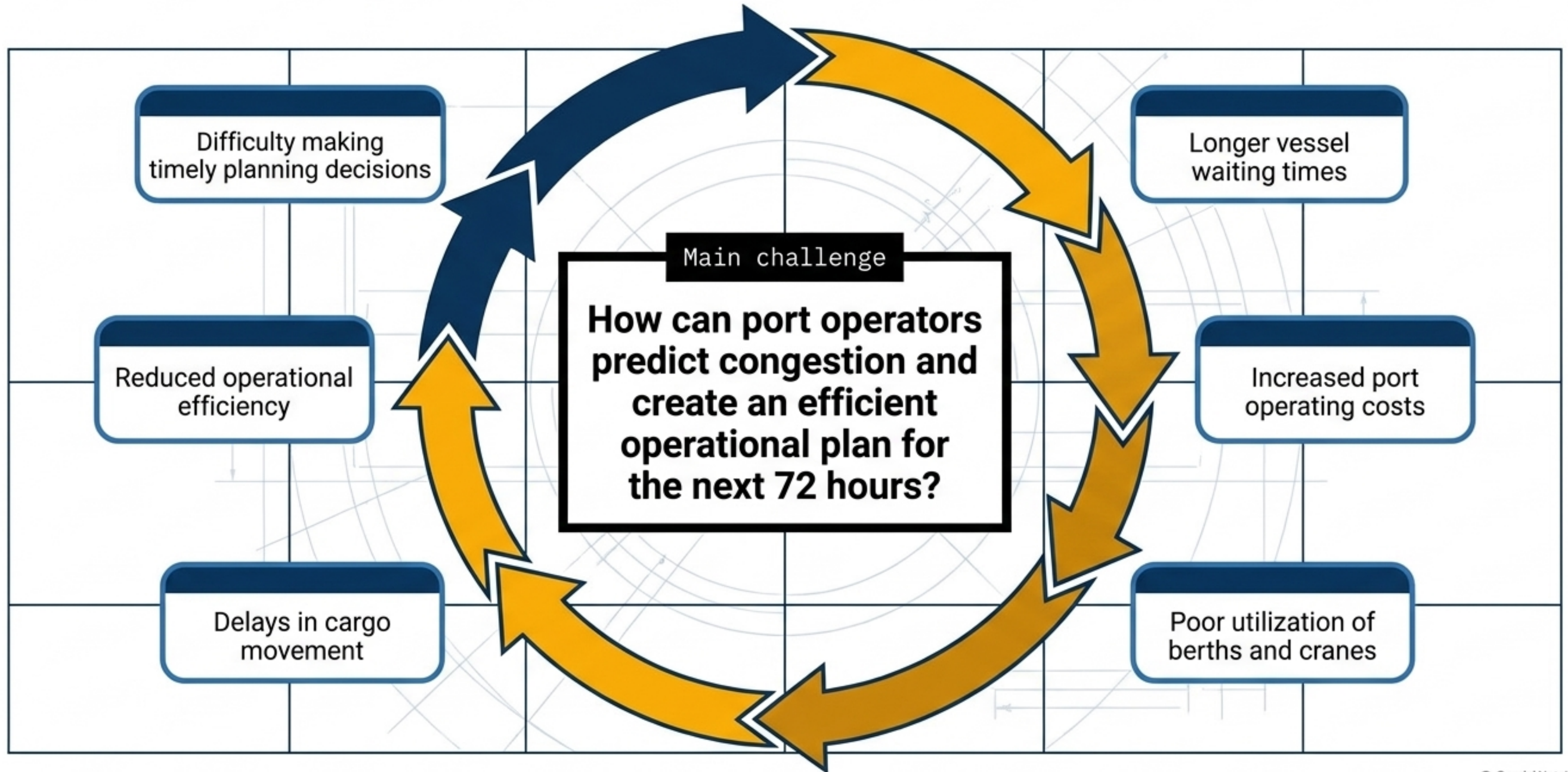
Hackathon: IBM BoB AI Innovation Hackathon 2026

Port Congestion and Inefficient Resource Planning



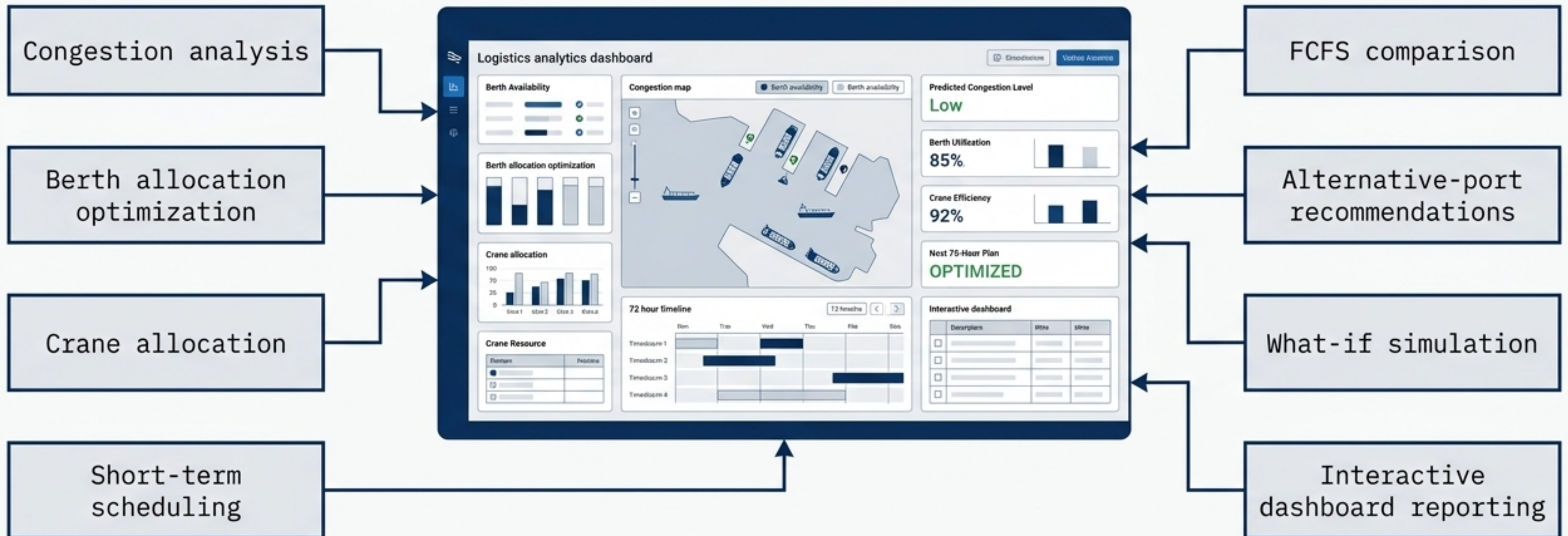
Port operations teams need an effective way to predict congestion and decide how to allocate berths and cranes efficiently.

Problem Impact



PortPilot AI

PortPilot AI is an AI-powered decision-support system that predicts port congestion and generates an explainable 72-hour operational plan.



Project Objectives

Predictive Intelligence

- Predict port congestion levels
- Identify the causes of congestion

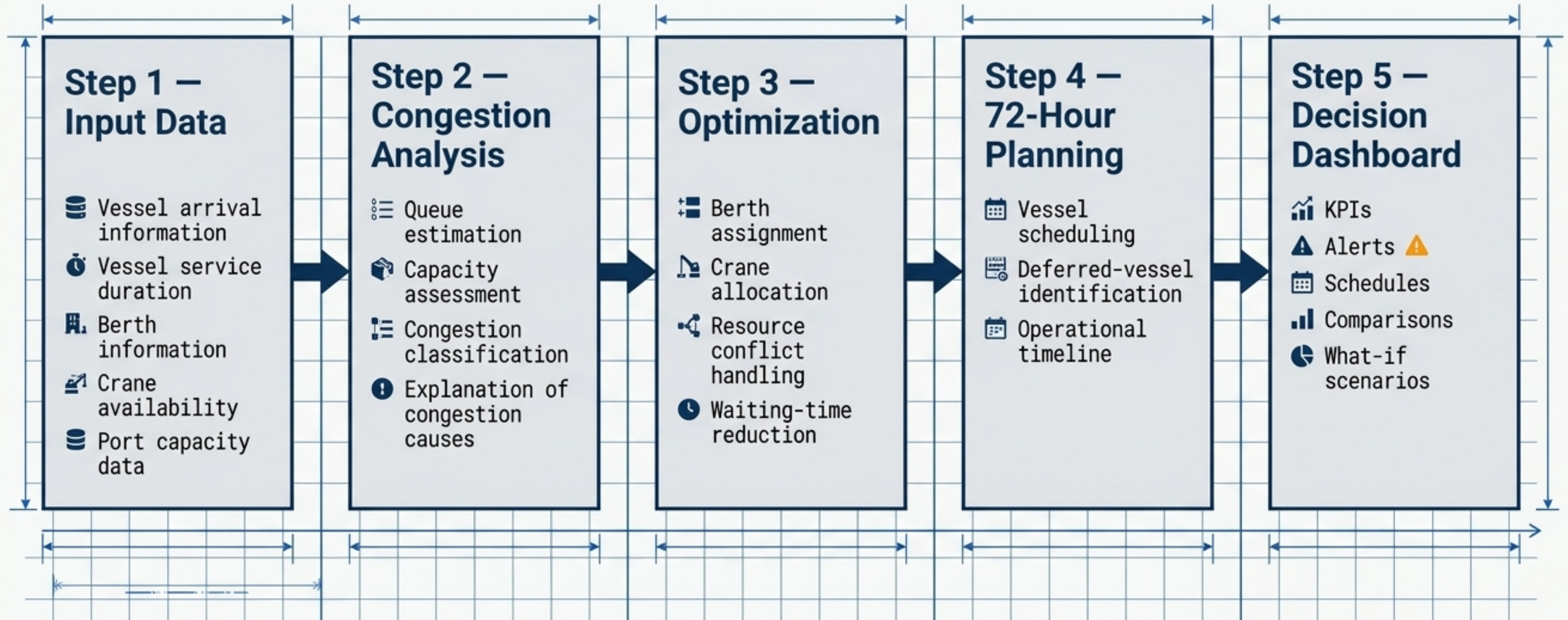
Resource Optimization

- Reduce vessel waiting time
- Optimize berth assignments
- Allocate cranes according to available capacity

Decision Support

- Generate a rolling 72-hour plan
- Compare optimized planning with FCFS
- Support alternative operational decisions
- Provide understandable recommendations

How PortPilot AI Works



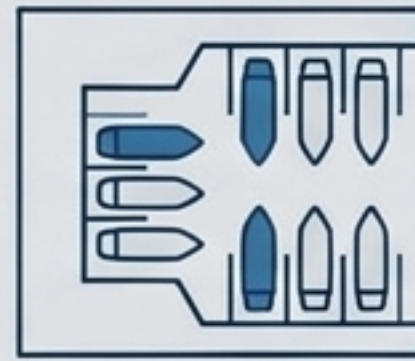
PortPilot AI Features



Congestion
Prediction



Explainable
Congestion Analysis



Berth Allocation
Optimization



Crane
Allocation



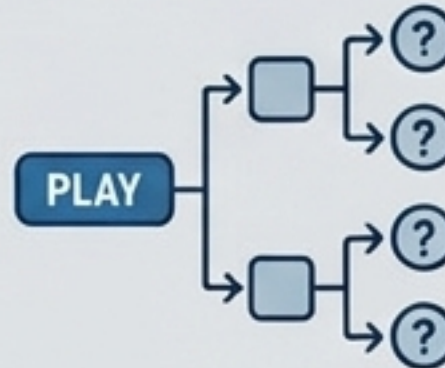
Rolling 72-Hour
Planning



FCFS Baseline
Comparison



Alternative-Port
Recommendations



What-If
Simulation

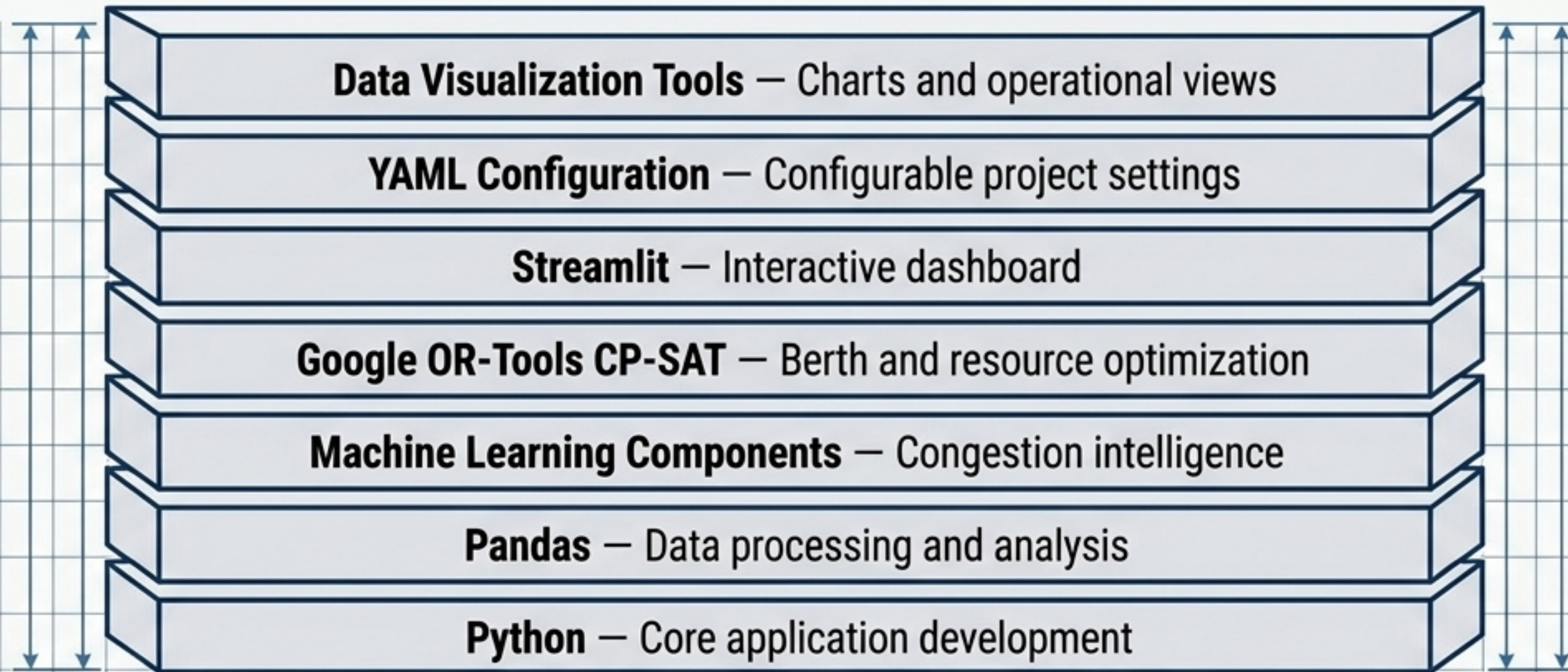


Operational
Alerts



Interactive Streamlit
Dashboard

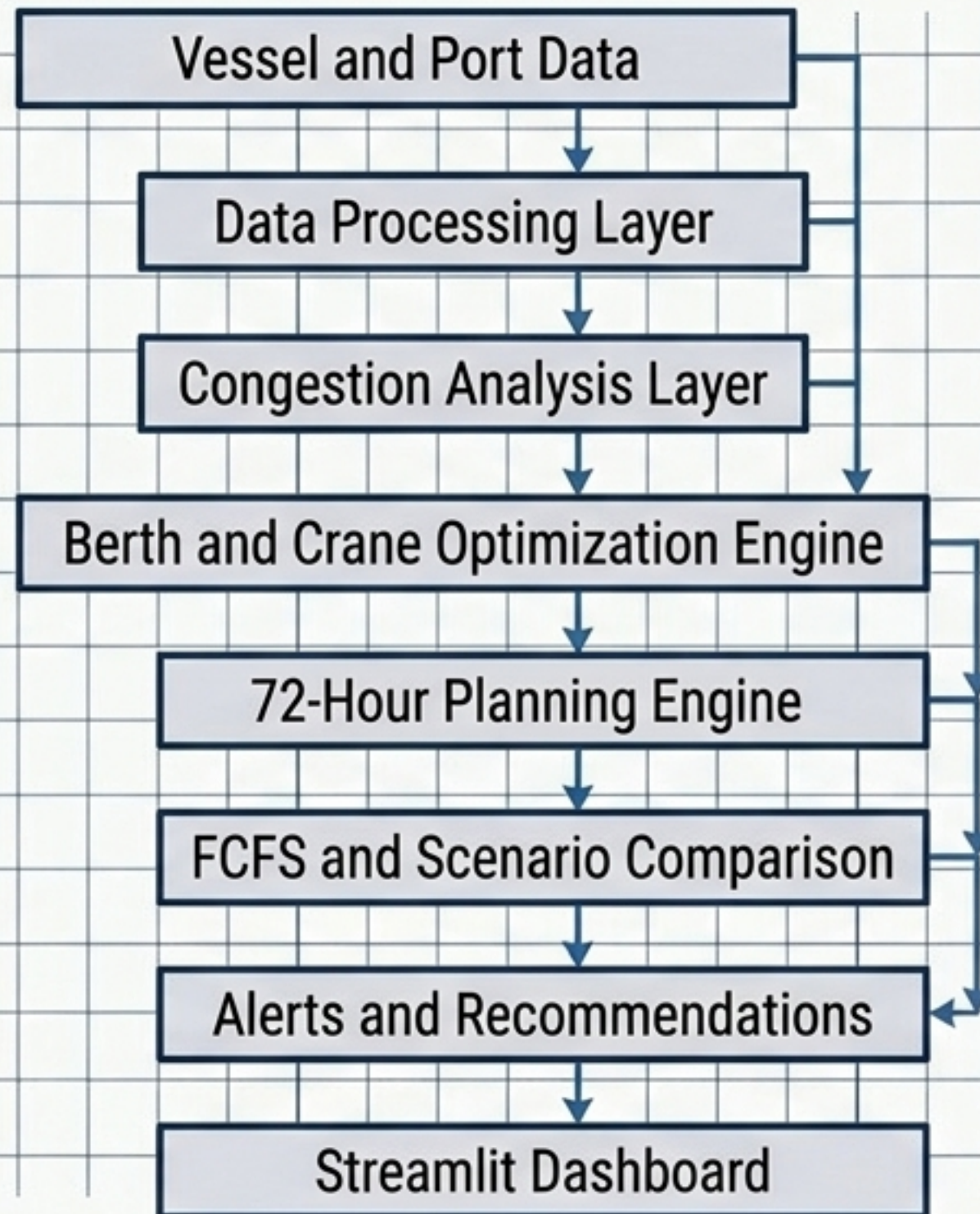
Technology Used



Core approach

AI identifies congestion conditions, while optimization determines how available resources should be used.

Architecture



Main design principle

Convert congestion predictions into feasible, explainable, and actionable operational plans.

Berth and Crane Allocation

The OR-Tools optimization engine considers:

Vessel arrival times

Vessel service duration

Berth availability

Scheduling horizon

Vessel-to-berth assignments

Crane availability

Maximum crane capacity

Resource conflicts

Optimization goals

Reduce
vessel
waiting
time

Improve
berth
utilization

Use
cranes
efficien-
tly

Handle
more
vessels
within
the
planning
window

Produce
a
feasible
schedule

Understanding Why Congestion Happens

PortPilot AI evaluates:

Vessel queue size, Available berths, Average service duration, Crane capacity, Vessel throughput, Resource utilization



LOW: Manageable operational pressure



MEDIUM: Increasing queue or resource pressure



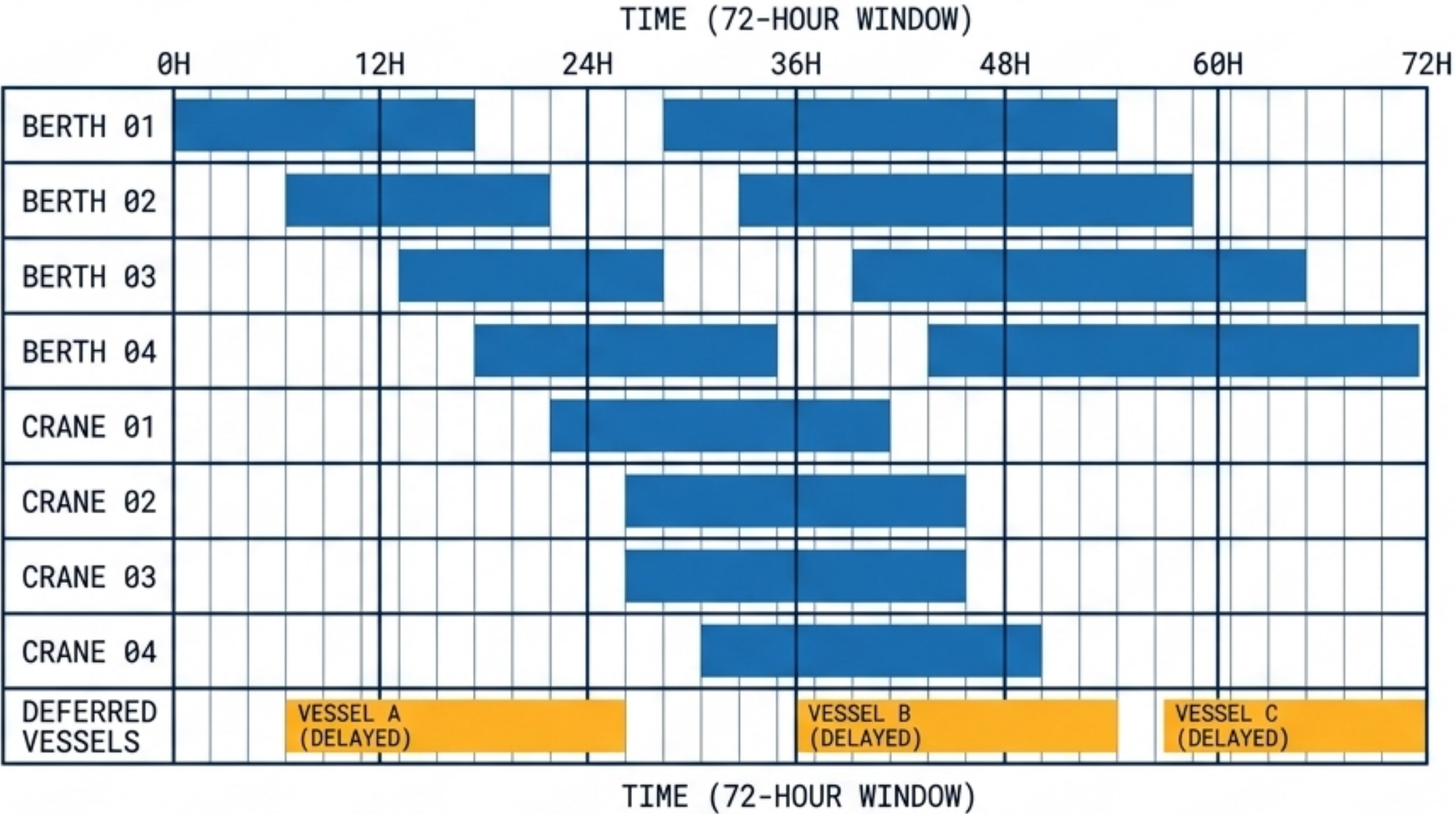
HIGH: Significant congestion and delay risk

Example explanation

"Congestion is increasing because vessel queue pressure is high while berth and crane capacity restrict the number of vessels that can be processed."

Short-Term Operational Planning

PortPilot AI generates a rolling 72-hour plan that includes:
Vessel arrival timeline
Vessel-to-berth assignments
Crane allocation
Expected waiting time
Scheduled service periods
Deferred vessels
Resource utilization



Benefit	Port operators can view upcoming operational pressure and prepare actions before congestion becomes more severe.
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Optimized Planning vs. First-Come, First-Served

